

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M.Tech)

Semester – II University Examination May 2010

M.Tech. (C.S.E.)

EC – 707 Advanced Optical Communication

Date: 21/06/2010, Monday

Time: 12:30 p.m to 03:30 p.m

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION -I

- Q-1(a) Conditions for the phenomenon of total internal reflection 3
- (b) What are the two important performance parameters of Photodiode 3
- (c) Define the following 2
- 1) Numerical Aperture 2) Acceptance Angle
- (d) The addition of the dopants to silica causes variations in refractive indices to the core and cladding of optical fiber, Explain the same. 3
- Q-2(a) Compare Unguided Optical communication system with radio and microwave systems 4
- (b) For an optical fiber with a core radius of $50\mu\text{m}$ operating at 850 nm . Calculate 1) refractive indices of the core and cladding materials with $\text{NA}=0.4$ and $\Delta=0.01$ 2) Critical angle 3) Acceptance Angle 4
- (c) Compare Single and Multimode Fibers. 4

OR

- Q-2(a) List out the advantages and disadvantages of Unguided optical communication systems 4

- (b) Explain Vapour phase axial deposition method for the fabrication of glass reform. 4
- (c) Explain scattering losses and core and cladding losses. 4
- Q-3(a) What are the main two transmission parameters related to unguided optical communication systems and explain them. 4
- (b) Compare the principal of operation and characteristics of RAPD with PIN. photo diode. 4
- (c) A PIN photo diode as a quantum efficiency of 65% at a wavelength of 0.85 μm . Calculate 4
 - 1) Its responsivity at 0.85 μm .
 - 2) Received Optical Power if the mean photo current is 10^{-6} A.

OR

- Q-3(a) List out the sources and detectors used in unguided optical comm. Systems and explain any one of them. 4
- (b) Explain the principle of operation, the structure and characteristic of the PIN photodiode. 6
- (c) State the three factors on which the response time of a photodiode with its associated output circuit depend? 2

SECTION -II

- Q-4(a) Define and explain the significance of the following terms related to the optical source. 6
 - 1) Radiance 2) Optical Confinement 3) Carrier Confinement 4) Quantum Efficiency 5) Emission Response Time 6) FMWH Spectral width
- (b) Write a short note on performance of WDM+EDFA systems. 5
- Q-5(a) Write down the public network application for optical fiber communication systems and explain them 7
- (b) Explain nearest and largest distance power budget calculation. 5

OR

Q-5(a) Explain the Civil, Consumer, Industrial and military application for optical fiber communication system. **7**

(b) Explain SONET / SDH Space Networks. **5**

Q-6(a) Define internal quantum efficiency, bulk recombination life time and spectral width of LED source. **7**

(b) Explain fabry parrot resonator cavity for a laser diode. **5**

OR

Q-6(a) Explain following with respect to laser diodes: **7**

1) The structure with neat and labeled sketch

2) Modes of the cavity

(b) Describe with a neat and labeled sketch the construction and characteristics of edge limiting LED. **5**
